

SEM NAN, Iran, Islamic Republic of

WMO: 407570

Lat: 35.5878N

Lon: 53.4216E

Elev: 1127

StdP: 88.50

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.1	-14.2	1.3	9.3	-12.1	1.5	7.8	8.2	4.6	6.4	5.9	0.7	330	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	40.0	18.1	38.8	17.9	37.6	17.6	20.7	35.6	19.7	35.0	19.0	34.2	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	12.3	29.7	13.9	11.4	29.1	12.7	10.5	28.5	65.1	35.5	61.6	35.0	58.9	34.4	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	7.8	6.4	-5.8	41.8	2.5	0.7	-7.6	42.4	-9.1	42.8	-10.5	43.2	-12.3	43.7		
(5)			-7.0	21.8	2.1	2.0	-8.5	23.3	-9.7	24.5	-10.9	25.6	-12.4	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	4.3	7.0	12.6	18.1	24.6	30.0	32.4	30.7	26.8	19.9	11.4	5.9
	DBStd	10.36	3.30	3.72	4.30	4.10	3.04	2.80	2.55	2.33	2.65	3.48	3.93	2.90
	HDD10.0	455	177	93	22	2	0	0	0	0	0	0	30	130
	HDD18.3	1605	434	316	184	51	1	0	0	0	0	24	209	385
	CDD10.0	3628	1	11	102	245	451	599	693	641	504	306	72	3
	CDD18.3	1737	0	0	6	44	193	349	435	383	254	72	1	0
	CDH23.3	23442	0	1	49	371	2035	5057	6777	5611	2951	586	4	0
	CDH26.7	13360	0	0	9	95	852	3007	4411	3376	1452	159	0	0
Wind	WSAvg	2.2	1.3	1.7	2.1	2.5	2.8	3.1	3.0	2.6	2.2	1.8	1.4	1.2
Precipitation	PrecAvg	206	24	26	32	30	17	8	10	6	7	7	17	21
	PrecMax	320	71	65	92	79	63	47	44	31	37	33	47	46
	PrecMin	107	8	3	5	1	0	0	1	0	0	0	0	3
	PrecStd	52	16	18	23	18	14	9	9	7	7	7	12	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	20.1	28.1	32.1	35.2	40.2	41.9	40.7	37.0	32.0	23.9	16.9
		MCWB	7.0	8.8	11.8	13.9	15.4	17.8	18.3	17.9	16.1	15.2	11.9	8.5
	2%	DB	13.2	16.8	24.1	29.2	34.0	38.6	40.5	39.1	35.6	30.0	21.5	14.2
		MCWB	6.2	7.2	10.7	13.2	15.1	17.4	18.2	18.1	16.2	14.3	11.5	7.0
	5%	DB	11.6	15.1	21.9	27.1	32.7	37.4	39.3	37.8	34.2	28.4	19.8	12.7
		MCWB	5.3	6.5	9.8	12.4	14.7	16.9	18.0	17.9	15.9	13.7	10.6	6.6
	10%	DB	9.9	13.3	19.9	25.2	31.1	36.2	38.1	36.6	33.1	26.9	17.8	11.0
		MCWB	4.5	6.0	9.0	11.8	14.4	16.6	18.0	17.6	15.8	13.1	9.4	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.1	12.8	15.2	18.0	21.3	22.2	21.6	19.7	17.2	15.4	9.4
		MCDB	13.0	14.8	21.9	26.1	32.1	35.0	35.3	36.3	32.0	28.3	19.9	14.4
	2%	WB	7.1	8.5	11.6	14.2	16.3	19.2	20.9	20.3	18.4	15.2	12.7	8.2
		MCDB	11.3	15.2	20.8	25.4	30.8	34.6	36.3	35.0	31.3	27.1	18.7	12.9
	5%	WB	5.9	7.4	10.8	13.5	15.6	18.3	20.0	19.5	17.4	14.4	11.2	7.3
		MCDB	10.5	13.6	20.1	24.3	29.9	33.6	35.7	34.3	31.1	26.2	17.6	11.4
	10%	WB	4.9	6.5	9.8	12.7	15.0	17.5	19.2	18.8	16.7	13.7	10.2	6.2
		MCDB	9.4	12.4	18.3	23.0	28.5	33.2	35.1	33.4	30.5	25.0	16.6	10.0
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.4	10.7	10.9	11.2	10.4	11.2	11.4	10.7	9.1	8.6
		MCDBR	10.2	11.2	12.3	11.8	11.8	11.6	11.0	11.7	12.2	11.9	10.6	10.0
		MCWBR	6.1	5.8	5.6	4.7	4.4	4.1	3.6	4.3	5.0	5.2	5.8	6.0
	5% WB	MCDBR	9.5	10.1	10.7	10.8	11.1	11.2	10.5	10.6	11.4	11.2	9.5	9.0
		MCWBR	5.9	5.7	5.4	4.7	4.4	4.6	4.2	4.2	5.1	5.3	5.8	5.7
Clear-Sky Solar Irradiance	taub	0.340	0.384	0.459	0.539	0.552	0.536	0.572	0.520	0.465	0.447	0.385	0.337	
	taud	2.218	2.076	1.880	1.741	1.739	1.799	1.740	1.853	1.958	1.976	2.110	2.246	
	Ebn at Noon	839	837	802	760	756	767	736	766	787	757	772	817	
	Edh at Noon	108	140	187	226	230	216	228	200	171	153	120	99	
All-Sky Solar Radiation	RadAvg	2.80	3.66	4.81	5.82	6.92	7.82	7.54	7.01	5.91	4.46	3.12	2.49	
	RadStd	0.20	0.31	0.37	0.44	0.31	0.36	0.25	0.28	0.18	0.18	0.23	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	-1.23	N/A	-0.58	-0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page